Peatland rewetting in Schleswig-Holstein

Where are the peat soils, how are they used, how much do they emit – and where would rewetting make the most sense?

Project report · status 26 October 2026 · Bonnie Winter

QGIS / open-source portfolio project · test area: Treene–Sorge lowland (≈ 304 km²) · developed with the help of an AI assistant (see 'Use of AI', chapter 2)

The project in one paragraph. Drained peat soils under farmland are among the largest greenhouse-gas sources in Schleswig-Holstein. The project maps them for a test area, estimates their emissions from national emission factors, tests how robust that estimate is, derives hydrological sub-catchments from a 1 m terrain model, and then uses integer optimisation to find where a CO2 saving target can be reached with the least collateral damage. The central result: if rewetting is planned field by field, it looks efficient but would in reality wet thousands of hectares of farmland on mineral soil; planned by whole sub-catchments, the same target is reached with almost none. A comparison with the Prussian survey of 1878–80 shows that most of today's peat was already wet land then – and that about a third of the former wet meadows have since lost most of their peat.

Key numbers

	Value	Comment
Test area	30,358 ha	hand-drawn polygon, 34 municipalities
Organic soils	15,952 ha (53 %)	Thünen map (Wittnebel et al. 2023)
of it farmed	11,102 ha	97 % permanent grassland, 278 ha cropland
Current emissions	≈ 354,000 t CO2-eq/yr	ESTIMATE: area × national factor
Saving if all were rewetted	≈ 293,000 t CO2-eq/yr	rewetted peat still emits 5.5 t/ha
Uncertainty (90 % range)	351,000 – 358,000 t … 144,000 – 487,000 t	fields deviating independently … the whole region deviating together
Mineral-soil farmland wetted	4,296 ha (best parcel plan) vs 52 ha (catchment plan)	target 75,000 t CO2-eq/yr (≈ ¼ of the potential)
Wet land 1878–80 still organic today	88 % (whole area), 92 % (block)	1880 map read automatically, 90 % ± 6 % accuracy (blind check)

Status

Step	Content	Status
00–04	inputs, overlay, emission estimate	done
01, 05	DGM1 download (355 tiles), hydrology with GRASS	done
07	optimisation: parcels vs. catchments	done
08	Monte Carlo uncertainty	done
09	atlas: 24 A4 pages, one per municipality	done
06	historical wet land 1878–80 vs. today (36 km² block)	done
–	README, GitHub repository (MIT licence)	done
06x–06v	automated reading of the 1880 map, whole test area, accuracy check	done

Contents

1	Question, context and study area
2	How the project is built (repository, configuration, software)
3	Data and licences
4	The analysis, step by step (00 → 09)
5	Main findings
6	Limitations of the whole approach
7	Problems encountered and how they were solved
8	Next steps
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Every analysis step in chapter 4 is described in the same five parts: **why** it is needed, **what the script does**, the **results**, the **decisions** taken and why, and its **limitations**. All emission figures are estimates and are labelled as such.

1 Question, context and study area

The problem

Peat forms where water-logged conditions stop plant material from decomposing. Drained for farming, the peat is exposed to oxygen and slowly 'burns' away: the stored carbon is released as CO2, the ground subsides, and nitrous oxide is emitted as well. Raising the water table again (**rewetting**) stops most of this – but it also ends conventional farming on the land affected.

For Schleswig-Holstein, the Land's Moor-Klimaschutz information (January 2024) gives about 152,800 ha of carbon-rich soils (≈ 10 % of the state), of which about 129,800 ha peat (≈ 103,200 ha fen, 26,600 ha bog) and 23,000 ha Anmoor; more than 81 % are used for agriculture, and these soils cause about 17 % of the state's greenhouse-gas emissions.

The questions

- Where are the peat soils, and how are they used?
- How much do they emit – and how certain is that number?
- Which areas form their own hydrological unit, so that raising the water there does not flood neighbouring land?
- Where can a given CO2 saving be reached with the least land, the fewest units – or the least collateral damage?
- How much of today's peat was already wet land in the 1870s – and how much has been lost since? (step 06)

Study area (test mode)

I ran the analysis on a **test area** first, as in my earlier NRW project: a hand-drawn polygon of 30,358 ha (≈ 304 km²) in the Treene–Sorge lowland between Friedrichstadt, Hollingstedt and the A7, covering parts of the districts Schleswig-Flensburg, Nordfriesland and Rendsburg-Eckernförde (34 municipalities). I chose it because the Thünen map shows one of the densest concentrations of fen and bog soils in the state there. Setting `TEST_MODE = False` in the configuration would run everything for the whole state.

> **Key point.** The test area is a drawn polygon, not a real river catchment. That is fine for developing and checking the method, but it cuts through natural drainage units at its edges – step 05 flags those cut-off sub-catchments.

2 How the project is built

Repository structure

```
peatland-rewetting-sh/
├─ config.py            all paths, field names, parameters, emission factors
├─ peatsh/common.py     shared helpers: AOI, clipping, GeoPackage I/O, reports
├─ scripts/             00 … 09, one numbered script per analysis step
├─ qgis/                QGIS project peatland_sh.qgz, styles, models
├─ data/raw/            downloads, never edited          (not in git)
├─ data/interim/        DGM1 GeoTIFFs, hydrology rasters  (not in git)
├─ data/processed/      peatland_sh.gpkg – all results     (not in git)
├─ outputs/             CSV tables, figures, atlas maps
└─ docs/reports/        one short markdown report per step
```

Three design rules

- **One configuration file.** Nothing is hard-coded in the scripts. File paths, field names, emission factors, thresholds and the optimisation target live in `config.py`. Changing the study area or a factor means changing one line.
- **One GeoPackage for all vector results.** Every script reads and writes layers in `data/processed/peatland_sh.gpkg`, named `<step>_<area>` (e.g. `emissions_test_catchment`).
- **A report after every step.** Each script writes a markdown report with its numbers, the parameters used and the limitations – so every result can be traced back. Scripts refuse to run with missing inputs (e.g. no cited emission factor) instead of using made-up values.

Software

Tool	Used for	Note
QGIS 3.4x LTR (OSGeo4W Shell)	Python environment for all scripts	GeoPandas 1.1.3, GDAL, SciPy 1.17 already included
GeoPandas / pyogrio / Shapely	reading, reprojecting, clipping, overlay, gap closing	steps 00–04, 06–08
GDAL (osgeo)	XYZ → GeoTIFF, mosaic, resampling, vectorising	steps 01, 05
GRASS r.watershed via qgis_process	flow direction, accumulation, sub-catchments	step 05
SciPy milp (HiGHS solver)	binary integer optimisation	step 07
NumPy / SciPy / Matplotlib	Monte Carlo, distribution fitting, figure	step 08
PyQGIS	layer styling, print layout, atlas export	step 09, runs inside QGIS

Why not conda / WhiteboxTools / PuLP as originally planned? Windows *Smart App Control* on my computer blocks unsigned programs. It blocked the Miniforge (conda) installer and would also block the WhiteboxTools and CBC solver executables. QGIS and its bundled Python, GDAL and GRASS are signed, so I moved the whole pipeline onto QGIS's own Python. I also dropped PuLP because version 4.0 changed its interface. I deliberately left Smart App Control on: switching it off cannot be undone without reinstalling Windows.

Use of AI

I used an AI assistant (Claude, Anthropic) throughout this project: to draft and debug the Python scripts, to explain methods and literature, to suggest approaches and to draft this report and the README. I defined the questions and the study area, downloaded and prepared the data, ran every script on my own computer, checked the results in QGIS, decided between the options for methods and parameters, and digitised the 1878–80 map by hand. All figures and maps in this report come from my own runs.

How to run

```
OSGeo4W Shell:
cd /d D:\PORTFOLIO\Peatlands\peatland-rewetting-sh\peatland-rewetting-sh
python scripts\00_check_inputs.py      (… then 01–08)
QGIS Python console (project qgis\peatland_sh.qgz open):
exec(open(r"…\scripts\09_export_maps_pyqgis.py", encoding="utf-8").read())
```

3 Data and licences

Dataset	Provider · licence	Used for	Notes
Aktualisierte Kulisse organischer Böden (Wittnebel, Frank, Tiemeyer 2023)	Thünen-Institut · CC BY 4.0	where the peat is; soil type	303,672 polygons, Germany; KAT_KURZ = soil category
Feldblöcke SH, reference date 1.1.2026	Land SH · dl-de/zero-2.0 (to confirm for 2026)	farmed land, arable vs. grassland	194,503 blocks; field HBN; CRS EPSG:4647
DGM1 (1 m terrain model)	LVermGeo SH, OpenGBD · CC BY 4.0	hydrology	XYZ text tiles 1×1 km; bulk-download index of 18,685 tiles
VG250 boundaries (state, districts, municipalities)	BKG · dl-de/by-2.0	statistics per municipality, atlas	fields GEN, AGS, SN_L, GF
Emission factors	Tiemeyer et al. 2020, Ecol. Indic. 109:105838	emission estimate, Monte Carlo	Table 5 (totals), Table 2 (ranges)
OpenStreetMap	© OSM contributors · ODbL	background maps	greyed in the atlas
Preußische Landesaufnahme 1878–80 (Chronologen WMS)	LVermGeo SH · CC BY 4.0	historical wet land (step 06)	layer Chronologen_1_1878-1880; digitised by hand
Zeichen-Erklärung für die Messtischblätter (1898)	Geobasis NRW reprint · free (dl-de/zero-2.0 recommended)	meaning of the 1880 map symbols	official key of the same survey series
Protected areas (FFH, NSG)	LfU SH · to check	not used yet	–

Coordinate reference system

Everything is processed in **EPSG:25832** (ETRS89 / UTM zone 32N), the system of the LVermGeo SH data. The Feldblöcke arrive in **EPSG:4647**: the same projection, but with the zone number '32' written in front of the easting (32 500 000 instead of 500 000). The scripts reproject automatically.

Attribution required on every map

© GeoBasis-DE/LVermGeo SH/CC BY 4.0 (DGM1) · © GeoBasis-DE/BKG (VG250) · Thünen-Institut (organic soils) · Land Schleswig-Holstein (Feldblöcke) · © OpenStreetMap contributors. The atlas prints these in its footer.

4 The analysis, step by step

The order below follows the logic of the analysis, not the script numbers: data → overlay → emissions → uncertainty → hydrology → optimisation → maps.

Step 00 – Check inputs

Why

Before any analysis it must be clear what is actually in each file: its coordinate system, its attribute fields and their values. Field names differ between datasets and years, so they are looked up, not assumed.

What the script does

- lists every input in `config.INPUTS`: exists?, number of features, CRS, geometry type
- shows every attribute field with five example values
- reports how many DGM1 GeoTIFFs are present

Results (test area)

All main inputs were found. The report gave the field mapping now in the configuration: soil type **KAT_KURZ** (NH fen, HH bog, MF former peat soil, DK_* peat under a mineral cover), land use **HBN** (Ackerland / Dauergrünland), parcel ID **FLIK**.

Decisions and why

- Count features from the file header (pyogrio.read_info) instead of loading 300,000 polygons.
- Do not use the Feldblöcke field FLAECHE: it is text with a decimal comma. Areas are recomputed from geometry everywhere.

Limitations

- Shows only a sample of 500 rows for example values – rare codes can be missed (step 03 therefore lists every land-use value it could not map).

Step 02 – Prepare inputs

Why

All layers must share one coordinate system and be cut to the study area before they can be overlaid.

What the script does

- loads the AOI polygon and writes it to the GeoPackage
- reads each input **only inside the AOI** (spatial mask – much faster for Germany-wide files)
- repairs invalid geometries, reprojects to EPSG:25832, clips, recomputes areas

Results (test area)

Layer	Features	Area (ha)
AOI	1	30,358
organic soils	1,470	15,952
Feldblöcke	7,315	21,922
municipalities	34	30,358

Decisions and why

- Thünen's own AREA_HA field is dropped and recomputed: it is wrong after clipping, and GeoPackage treats area_ha and AREA_HA as the same field name.
- keep_geom_type when clipping, so stray lines/points from the cut don't end up in polygon layers.

Limitations

- Protected areas are not yet included.

Step 03 – Organic soils × agricultural land use

Why

Emissions depend on how the peat is used. The overlay answers: how much peat lies under arable land, how much under grassland, and in which municipality?

What the script does

- maps HBN codes to classes: Ackerland → cropland, Dauergrünland → grassland
- intersects Feldblöcke with organic soils, then with municipalities
- summarises hectares by soil type × land use and per municipality

Results (test area)

11,102 ha of the 15,952 ha organic soils (70 %) are farmed – **97 % of it permanent grassland** (10,824 ha), only 278 ha cropland. 104 'non-agricultural eligible areas' and one permanent crop were excluded.

Soil class	Cropland ha	Grassland ha
NH – fen	79	4,197
MF – former peat soil	112	1,773
HH – bog	3	1,693
DK_FL_NH – fen, shallow mineral cover	19	1,684
DK_MA_NH – fen, thick mineral cover	57	1,085
DK_FL_HH – bog, shallow mineral cover	8	392

Decisions and why

- Land use from the Feldblöcke (agricultural reference parcels), not from ATKIS: they are the official, current (2026) record of what is farmed.
- Very small overlay slivers are dropped.

Limitations

- A Feldblock is not an ownership parcel – one block can belong to several owners or farmers.

Step 04 – Emission estimate

Why

The core quantity of the project: how much greenhouse gas do the farmed peat soils emit, and how much would rewetting save?

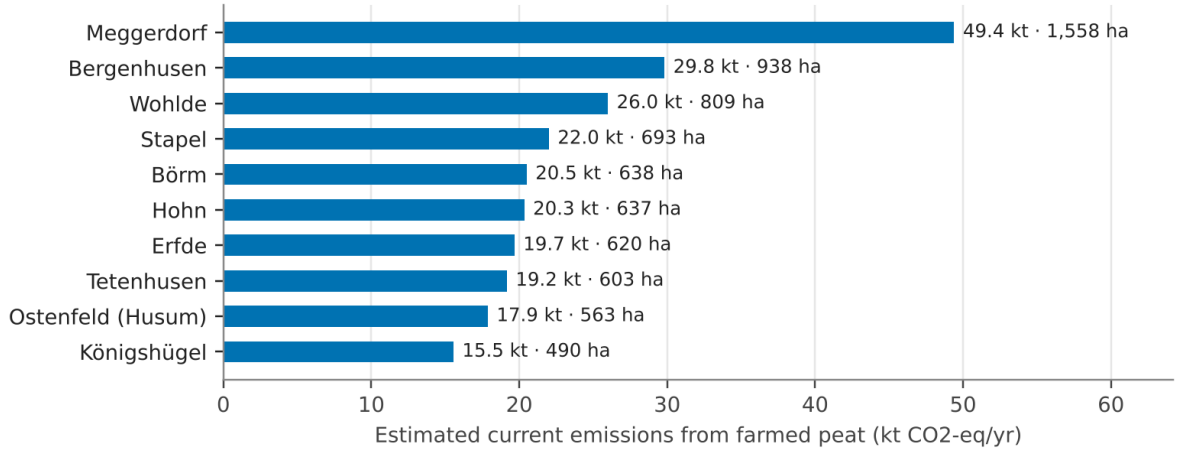
What the script does

- assigns each overlay piece the emission factor of its land-use class
- emissions = area × factor; saving = area × (factor − factor for rewetted peat)
- sums per municipality

> **Formula.** $E = \Sigma_c A_c \times EF_c$ · $saving = \Sigma_c A_c \times (EF_c - EF_{rewetted})$ with A = farmed peat area (ha), EF = emission factor, c = cropland / grassland.

Results (test area)

≈ 354,000 t CO2-eq/yr current emissions; **≈ 293,000 t** could be saved if all farmed peat were rewetted (≈ 61,000 t remain, mainly methane from wet peat). Meggerdorf alone accounts for ≈ 49,000 t.



Estimated current emissions by municipality (top 10). Because almost everything is grassland with one factor, the ranking simply follows the farmed peat area.

Class	Factor (t CO2-eq/ha/yr)	Source
cropland on organic soil	40.4	Tiemeyer et al. 2020, Table 5
grassland on organic soil	31.7	Tiemeyer et al. 2020, Table 5
rewetted organic soil	5.5	Tiemeyer et al. 2020, Table 5

Decisions and why

- **Implied national emission factors** from the German greenhouse-gas inventory method: they include CO2 (incl. dissolved organic carbon), methane from soil and ditches and nitrous oxide, converted with the IPCC AR4 warming potentials, and they already contain the undrained share of each land use – so they can be multiplied with area directly.
- The same factor is applied to all soil classes incl. former peat soils and covered peat: the paper found no emission difference between 'true peat' and lower-carbon organic soils at German farm sites (sect. 4.1.1).
- The script refuses to run without cited factors.

Limitations

- National averages: the actual water table, peat thickness and management of each field are ignored.
- Parts of the test area are in wet-grassland schemes; there the estimate is probably too high.

Step 08 – Monte Carlo: how certain is the estimate?

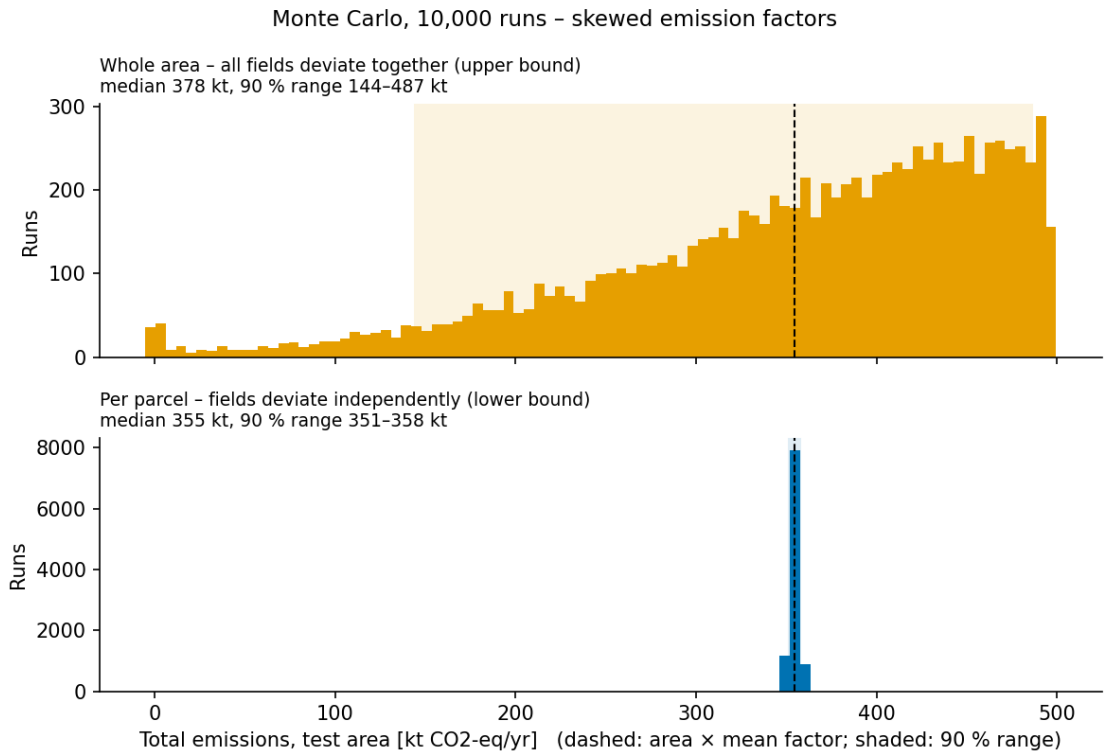
Why

A single number like 354,000 t hides its uncertainty. The question is how wide the plausible range is – and what drives it.

What the script does

- draws emission factors 10,000 times from a distribution fitted to the paper's numbers
- recomputes total emissions and savings for every draw
- runs two scenarios: **whole area** (one draw for all fields) and **per parcel** (an independent draw for each of 4,438 parcel/land-use units)

Results (test area)



Monte Carlo, 10,000 runs. Top: all fields deviate together. Bottom: fields deviate independently. Same x-axis, so the widths compare directly.

Scenario	Emissions median	90 % range	Savings median	90 % range
whole area (upper bound)	377,750	143,853 – 486,858	313,018	76,205 – 447,116
per parcel (lower bound)	354,681	350,943 – 358,406	293,210	289,232 – 297,117

Decisions and why

- **The key insight.** The paper's ranges (Table 2) are percentiles over all grid points: they describe how much *single hectares* differ, mainly because of their water level – not how uncertain the *average* over a large area is. My first version, which applied one random factor to all fields, therefore greatly overstated the uncertainty.
- **Lopsided distribution.** For drained land, most hectares emit close to a ceiling (CO2 stops rising once a field is deeply drained) with a long tail towards wetter fields. A two-piece normal distribution is fitted so that its mean equals the inventory factor exactly and its 2.5/97.5 % points match the paper within ≈ 0.7 t C/ha.
- **Physical floor.** Even a fully wet field cannot emit less than the lower limit of the paper's CO2 response curve (−0.93 t C/ha/yr); without this cap the tail produced impossible values.
- **Two bounding scenarios** instead of one guess about how strongly fields are correlated.

Limitations

- Only emission-factor variability is included – not mapping errors, the uncertainty of the response functions themselves, or the choice of warming potentials.
- Reality lies between the two scenarios: neighbouring fields share ditches and pumps.

Key point. The emission factors themselves don't make the estimate uncertain – the unknown **regional water level** does. If the whole region deviates from the national average, the range is −60 % / +35 %; if fields deviate independently, it is ±1 %. Local water-level data would narrow this down.

Step 01 – DGM1 terrain model

Why

Hydrology needs a detailed terrain model; in flat fen landscapes a few centimetres decide where water flows. The 1 m DGM1 of Schleswig-Holstein is open data.

What the script does

- works out which 1×1 km tiles touch the AOI (355)
- downloads them via the portal's official **bulk-download index** (a GeoJSON with a direct link per tile), one at a time with a 1-second pause
- converts each XYZ text tile (1,000,000 lines of x y z) straight into a compressed 1 m GeoTIFF – the 28 MB text files are never stored; ≈ 0.6 s per tile
- records each tile's survey year and writes a tile layer (present / missing / year)

Results (test area)

All **355 tiles** complete. The survey years form clear blocks: west and centre mostly **2025** (205 tiles), a northern strip **2024** (39), the east (easting ≥ 528 km) **2005/2006** (106), plus 2017 (1) and 2021 (4).

Decisions and why

- The interactive download client allows only one tile per order; the bulk index made the download automatic and reproducible – and it is the sanctioned route for mass downloads.
- HTTPS certificates are checked against the Windows certificate store (package truststore); certificate checking was never switched off.
- Safe to re-run: finished tiles are skipped; failures are listed.

Limitations

- Tiles almost 20 years apart meet at easting ≈ 528 km. Drained peat subsides by roughly 1 cm per year, so steps of centimetres to decimetres can appear at tile edges – relevant in flat terrain.

Step 05 – Hydrology with GRASS r.watershed

Why

Rewetting raises the water level in a whole hydrological unit, not in a single field. To plan realistically, those units are needed: sub-catchments and the streams that connect them.

What the script does

- mosaics the 355 tiles and resamples them to a 5 m working grid
- runs GRASS **r.watershed** through qgis_process: flow direction (D8), flow accumulation, streams, sub-catchments of at least 50 ha (20,000 cells)
- vectorises the sub-catchments, computes their share of organic soil and flags catchments touching the AOI/DEM edge as incomplete

Results (test area)

	Run 1 (mean)	Run 2 (min, used)
sub-catchments	393	367
incomplete (touch the edge)	80	80
complete and ≥ 50 % organic soil	207	196 (14,741 ha; 12,119 ha peat)

Visual check: the catchment boundaries run largely along dykes, dams and road embankments – realistic for a polder landscape, where these are the true water divides.

Decisions and why

- **r.watershed** routes water through depressions with a least-cost search, so no 'fill sinks' step is needed. Filling would create large artificial flats in fen terrain.
- **Minimum instead of mean** when resampling 1 m → 5 m. Ditches are only 1–3 m wide; averaging smoothed them away and streams wandered across fields in stair-steps. Taking the lowest 1 m cell keeps ditch bottoms connected – the streams now follow the ditch pattern.
- Outputs are named after the method (stream_min.tif …), so both runs can be compared in QGIS.

Limitations

- **Pumping stations.** The polders are drained by ditches and pumps that lift water into the embanked Alte Sorge. A terrain model cannot see pumps, culverts or pipe drains, so the main river is not recognised as the main drain.
- The real drainage units are the pumping-station catchments of the water and soil associations (Wasser- und Bodenverbände); the DEM catchments approximate them.
- Possible improvement: 'stream burning' (GRASS r.carve) with the mapped ditch network.

Step 07 – Optimisation: parcels vs. catchments

Why

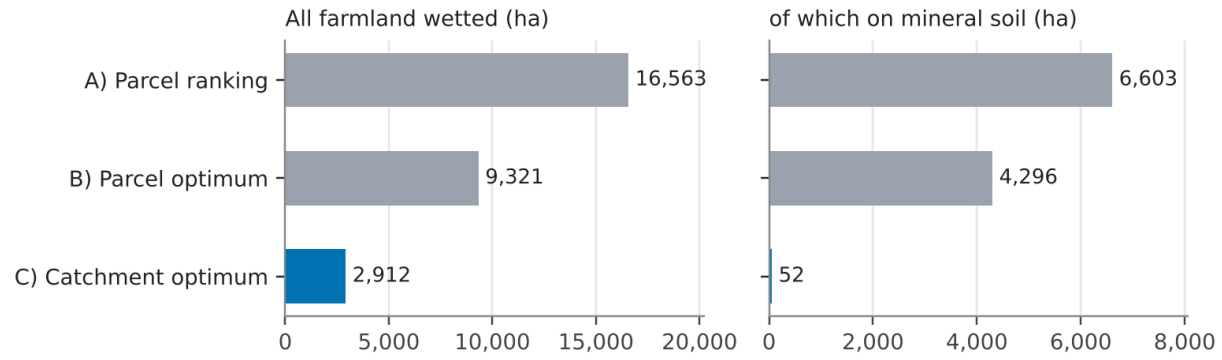
A ranking ('best fields first') is the intuitive way to choose areas. The question is whether an optimisation does better – and what changes when hydrology is taken into account.

What the script does

- **A) Parcel ranking:** Feldblöcke sorted by saving per ha, taken until the target is reached
- **B) Parcel optimum:** binary integer programme over 3,411 parcels – least area that reaches the target
- **C) Catchment optimum:** the same over 272 complete sub-catchments – a catchment is chosen whole; the objective is to wet as little farmland on **mineral** soil as possible
- for A and B it also counts what would really get wet: all farmland in every catchment that contains a chosen parcel

Integer programme. Choose $x_i \in \{0, 1\}$ for every unit i (parcel or catchment): minimise $\Sigma\ \text{cost}_i \cdot x_i$ subject to $\Sigma\ \text{saving}_i \cdot x_i \geq \text{target}$. Cost = area (B) or mineral-soil farmland in the catchment (C).

Results (test area)



All three approaches save ≈75,000 t CO2-eq/yr. Parcel approaches: farmland in every catchment that contains a chosen parcel.

Approach	Units	Peat rewetted ha	Saving t/yr	Catchments	Farmland wetted ha	on mineral soil ha
A) ranking	914 parcels	2,791	75,037	230	16,563*	6,603*
B) parcel optimum	808 parcels	2,789	75,001	119	9,321*	4,296*
C) catchment optimum	82 catchments	2,860	75,007	82	2,912	52

* if the water level is raised in every catchment containing a selected parcel – which is what rewetting physically does.

Decisions and why

- Target 75,000 t CO2-eq/yr ≈ a quarter of the potential: large enough that the solver has to choose.
- Solver: SciPy milp (HiGHS) – no external program, all three models solved to proven optimality in seconds.
- Without hydrology, A and B are almost identical (2,791 vs 2,789 ha): with only two emission factors nearly every hectare saves the same, so optimisation cannot beat a ranking – an important result in itself.
- Each parcel is assigned to the catchment containing its interior point, so parcels touching a boundary are not counted twice (this corrected A and B by ≈ 10–15 %).
- Objective switchable in config.py: non_peat_farmland (default), affected_farmland, count.

Limitations

- Uses the DEM catchments (see step 05). With pumping-station catchments the result could change.
- No costs, ownership, protected areas or farmers' willingness yet.
- Savings are estimates with the uncertainty from step 08.

Key point. Planned field by field, rewetting looks efficient but would wet 4,000–6,600 ha of farmland on mineral soil that emits nothing. Planned by sub-catchment, the same 75,000 t are reached with 52 ha – about 80 times less collateral damage. Hydrology turns the optimisation from a formality into the core of the analysis.

Step 06 – Wet land 1878–80 vs. organic soils today

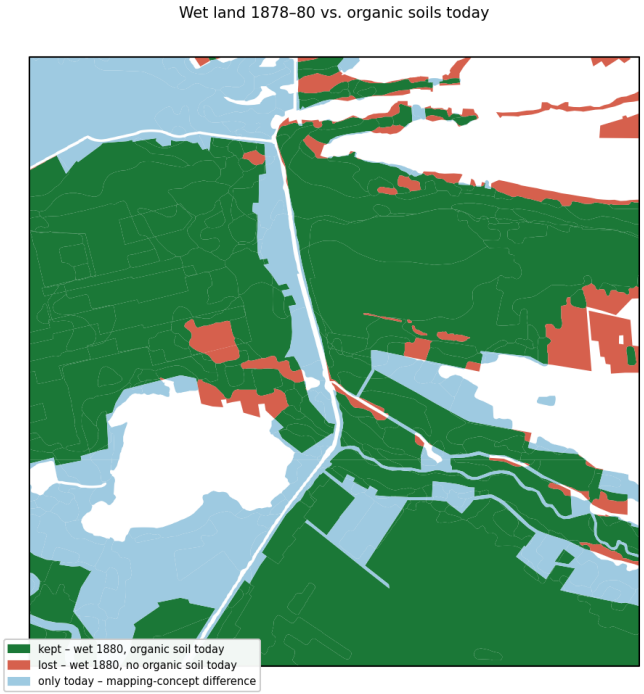
Why

Today's soil map shows where peat is now. The first Prussian survey of 1878–80 shows where the land was wet 150 years ago. Comparing the two shows how stable the peat landscape is, where peat has been lost, and how far drainage had already progressed by 1880.

What the script does

- **manual part (QGIS):** I digitised the wet signatures of the 1878–80 sheets (WMS Chronologen) by hand at 1:2,500–1:5,000 in a 6 × 6 km block around Meggerdorf (36 km²), recording the **legend class** the map shows rather than an interpretation; the checked area is stored separately as `digitised_extent`
- checks every polygon's class and turns the free-text evidence (e.g. 'torfstich, tufts, long dashes') into codes (bruch, torfstich, name)
- clips to the checked area and closes gaps of up to 30 m between neighbouring polygons (roads, ditches)
- overlays with today's organic soils: **kept** (wet then, organic now), **lost** (wet then, no organic soil now), **only today** (organic now, no wet signature then)
- reports how much of each difference lies within 25 m of a boundary (possible position error)

Results (test area)



Green: wet in 1878–80 and organic soil today. Red: wet then, no organic soil today. Light blue: organic soil today without a wet signature in 1880. Black frame: the 36 km² block that was checked.

Category	ha	Meaning
kept	2,007	92 % of the 1880 wet land
lost	175	≈ ⅓ of it within 25 m of a boundary – real loss ≈ 120–175 ha
only today	721	mostly fen (NH 423 ha) shown as pasture, dry meadow or arable in 1880

1880 class	Area ha	Kept %	Main soil type today (kept area)
Moor	1,086	95	Hochmoor 787 ha, Moorfolgeboden 119 ha
nasse Wiese	1,067	91	Niedermoor 465 ha, **Moorfolgeboden 336 ha**
Heide	29	6	– (mostly no organic soil today)

Decisions and why

- **Legend from the same survey series.** A 1933 legend first suggested one meadow class; the official *Zeichen-Erklärung für die Messtischblätter* (1898) separates nasse Wiese, trockene Wiese, Hutung/Viehweide, nasser Boden, Heide and Sumpf/Bruch/Moor. It explained why two different dash patterns appeared on the map, and I rebuilt the classes on it.
- **Record what the map shows, interpret later.** I digitised the classes Moor (peat certain), nasser Boden and nasse Wiese (likely) and Heide in the lowland (possible). I left out pasture, dry meadow, sand, arable land and villages.
- **Recurring Torfstich makes Moor**; a single peat cutting is only noted, so one symbol cannot turn a whole meadow into moor.
- **A separate extent layer**, so unchecked land is not mistaken for 'dry in 1880'.
- **Gap closing tested:** 15 m vs. 30 m changes every category by ≤ 1 %; I used 30 m.

Limitations

- Map signatures (land use and wetness) are compared with a soil map – different concepts. 'Only today' is therefore no gain of peat but fen that was already drained in 1880.
- Hand digitising, the positional accuracy of the 1880 survey and a key ~20 years younger than the sheets add uncertainty.
- One 36 km² block, which I chose for its mix of bog, fen and geest; not representative of the whole state.

Key point. Peat in this lowland has been remarkably stable in extent: 92 % of the land that was wet in 1878–80 is still organic soil. But it has thinned: about a third of the former wet meadows are now **Moorfolgeboden**, soils whose peat has largely gone – the long-term result of the drainage that steps 04 and 07 are about.

Steps 06x–06c – Reading the 1880 map automatically (YOLO)

Why

Hand digitising the 36 km² block took me several hours; the whole state is more than 400 times larger. I wanted to know whether the same classes can be derived automatically – with a small object-detection model for the map symbols and the same rules I used by hand – trained and run entirely on my own computer.

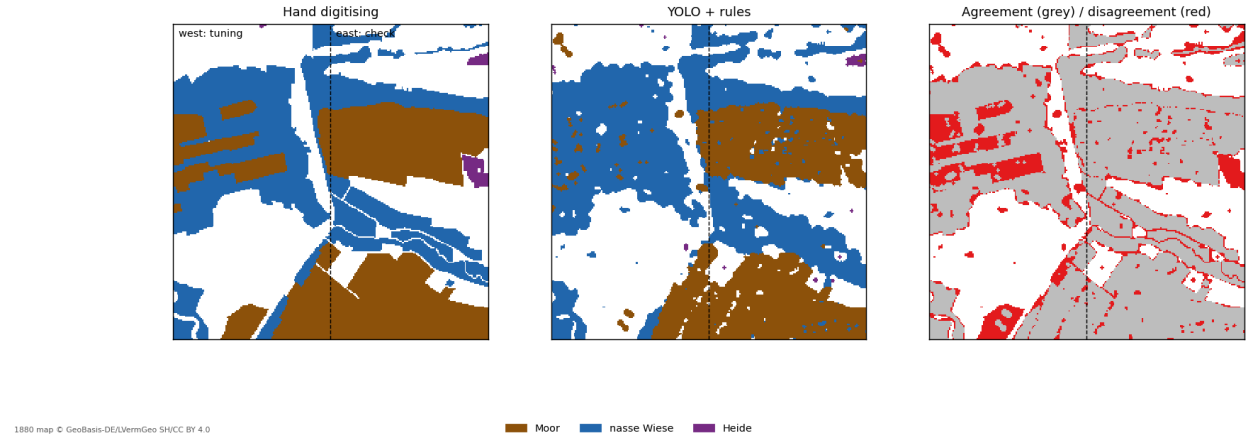
What the script does

- **06x:** the 1 m GeoTIFF of the block (exported from the Chronologen WCS) is cut into 361 tiles of 320 m, enlarged to 640 px, so a tuft becomes about 25 px instead of 12
- **labelling (QGIS):** I marked every tuft (point) and every peat cutting (Torfstich, rectangle) in 32 tiles – 400 tufts, 25 chains – including 11 tiles without symbols (dense dashes, lettering, a village)
- **06y:** labels → YOLO format (tuft points → 16 m boxes); split 24 / 4 / 4 tiles, with tiles that share an edge kept in the same set and every set containing moor, meadow and empty tiles
- **training:** Ultralytics YOLO26n on the CPU (about 15 minutes), stopped early at epoch 124
- **06z:** point-based check – a symbol counts as found if a detection lies within 8 m
- **06b:** detection on all 361 tiles, duplicates merged, converted to map coordinates
- **06c:** dash filter + tufts + Torfstich → classes on a 25 m grid; thresholds tuned on the west half of the block, result checked on the east half

Results (test area)

Symbol detection (4 test tiles, conf ≥ 0.20)	Recall	Precision*
tuft (53 labelled)	0.75	0.59
Torfstich (4 labelled)	3 of 4	0.50

* before duplicates are merged; two boxes on one tuft count as one hit and one false hit.



Left: my hand digitising. Middle: YOLO detections + rules. Right: agreement (grey) and disagreement (red). Dashed line: west half used for tuning, east half for checking.

Area	Agreement	Cohen's κ	IoU Moor	IoU nasse Wiese
west (tuning)	84 %	0.74	–	–
east (not used for tuning)	**90 %**	**0.86**	0.91	0.76

Decisions and why

- **Tiles of 320 m enlarged ×2.** YOLO looks at the image in steps of 8 px; a 12 px tuft is barely more than one step. Enlarging adds no detail but gives the network enough pixels.
- **Less scale augmentation (scale = 0.2).** The default shrinks training images by up to 50 %, which turns tufts into specks that never occur on the real map.
- **A point-based check instead of mAP alone.** mAP50 (0.52 for tufts) counts a box 5 m off as a miss. For counting tufts per 75 m window that is irrelevant; the point check showed the model finds 86 % of tufts in the validation tiles – with low confidence, so the threshold was lowered to 0.20.
- **Dashes without machine learning.** Flat ink marks (≤ 3 m high, 5–40 m long) are the dashes of Bruch and nasse Wiese; dotted pasture rows, lettering, buildings and field lines fall outside these limits. Dash density alone separates wet from not-wet land with an AUC of 0.97.
- **The same rules as by hand**, including 'heath in a moor': heath tufts near recurring Torfstich, or joined to a larger moor, become Moor. I added this rule after a first run classified the drained Königsmoor as heath (κ 0.76 → 0.85); it comes from my digitising rules, not from the check data.
- **Honest evaluation.** Thresholds are tuned on the west half only; the east half is the check. The YOLO test tiles were evaluated once, at the end.

Limitations

- The model was trained on tiles from the same block it is evaluated on (different tiles, same sheet and engraver). Other sheets need spot checks before it is applied there.
- Moor names are not detected; the small Heide class (29 ha) is not reproduced.
- Nasse Wiese is drawn slightly too wide (78 ha in the east half where I drew nothing), mostly into pasture with fairly dense dash rows.
- Only 23 Torfstich chains for training – that class is the weakest part of the model.

> **Key point.** A small model trained on 32 hand-labelled tiles, combined with simple image analysis for the dashes and my own digitising rules, reads the 1880 map almost as I did by hand: 90 % agreement on land it was not tuned on.

Applying it to the whole test area (06x–06c aoi, 06, 06v)

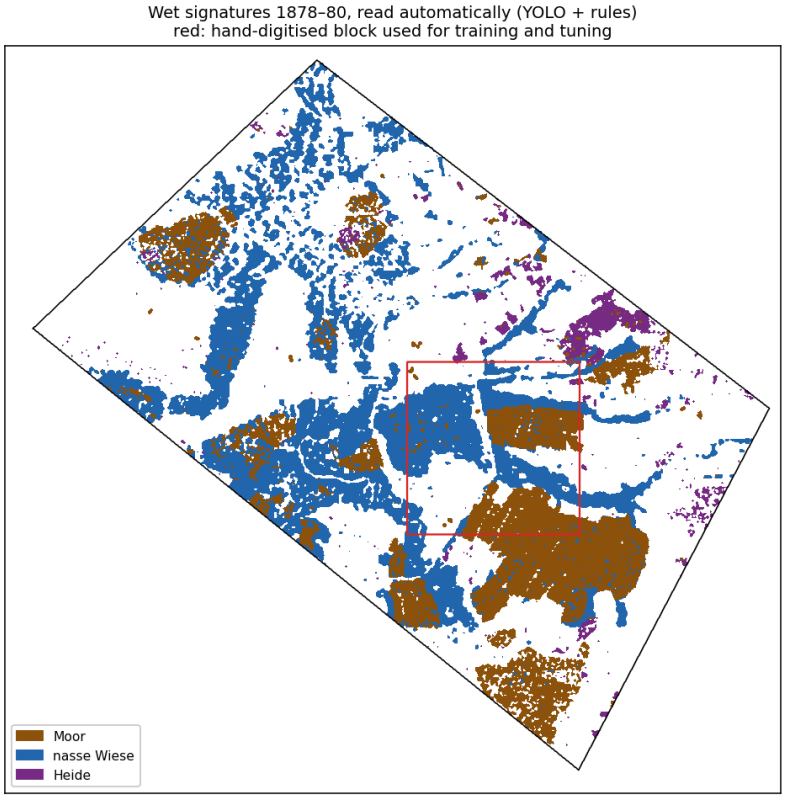
Why

The point of automating the map reading is scale. I applied the model and the thresholds tuned on the block, without re-tuning, to the whole test area (304 km², about eight times the block) – and then checked the result blindly, because outside the block there is no hand-digitised reference.

What the script does

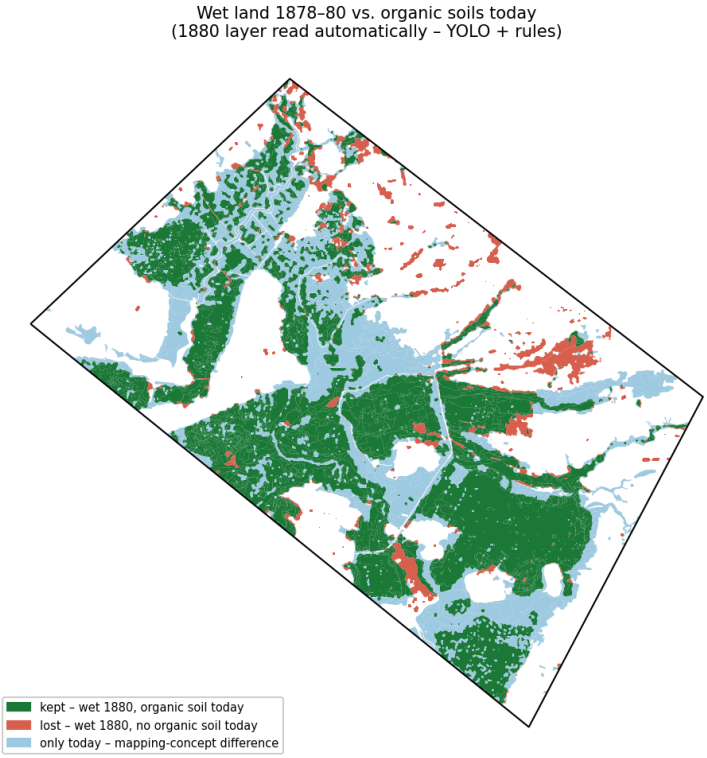
- exported the 1880 map for the whole test area from the WCS (27,000 × 26,000 px at 1 m) and cut it into 3,119 tiles inside the AOI
- detection on all tiles in about 20 minutes on a laptop CPU: 11,943 tufts, 771 Torfstich
- 12 random spot-check tiles outside the block with the detections drawn, which I inspected
- rules applied with the block thresholds; step 06 repeated for the whole area
- **06v:** 100 random points outside the block, 25 per mapped class; I classified each from the 1880 map without seeing the prediction; area-weighted accuracy after Olofsson et al. (2014)

Results (test area)



1880 classes for the whole test area, read automatically. Red frame: the hand-digitised block used for training and tuning.

Class (outside the block)	Mapped ha	User's acc.	Producer's acc.	Estimated ha ± 95 %
Moor	3,478	0.92	0.86	3,714 ± 726
nasse Wiese	5,665	**0.72**	0.86	4,754 ± 1,668
Heide	752	0.84	0.69	910 ± 393
none	16,862	0.96	0.93	17,381 ± 1,605
Overall accuracy		**90 % ± 6 %**		



Wet land 1878–80 (automated) vs. organic soils today, whole test area.

	Block (hand, 36 km²)	Whole test area (automated)
wet land 1880	2,182 ha	11,385 ha
still organic soil today	92 %	88 % (10,020 ha)
lost	175 ha	1,365 ha (upper bound)
1880 Moor → today's raised bog (HH)	73 %	83 % of kept
1880 nasse Wiese → Moorfolgeboden	31 %	20 % of kept
organic today, not wet in 1880	721 ha	5,932 ha

Decisions and why

- **No re-tuning outside the block.** The thresholds come from the block; the whole-area result is therefore a real test of transfer, and the 100 points are an independent check.
- **Two rule fixes after the first whole-area run.** A dry heath near Friedrichsneuland came out as Moor: tuft base lines were counted as dashes, and the heath-in-a-moor rule spread over the whole heath from a few moor-like cells. Base lines within 10 m of a tuft are now ignored, and heath only joins a moor at least as large as itself. On the block this changes nothing (κ 0.85 → 0.86). I made these fixes before drawing the 100 check points.
- **Automated Heide left out of the 1880 wet land**: heath away from any moor is mostly geest heath, not evidence of peat (lost 2,017 → 1,365 ha).
- **The check points were not used for any further tuning**, so the 90 % stays an honest figure.

Limitations

- Nasse Wiese is over-mapped by roughly 900 ha (user's accuracy 0.72), mostly into dense pasture rows and dry meadow – the 'lost' area is therefore an upper bound.
- Peat cuttings at steep angles and larger tufts drawn in a different style are often missed (spot checks).
- Land around peat cuttings becomes ~125 m squares, not whole fields as I would draw by hand.
- 25 points per class give wide confidence intervals, especially for nasse Wiese.

Key point. Applied without re-tuning to an area eight times larger, the method reaches 90 % ± 6 % accuracy in a blind check. The main historical result holds at this scale: most of today's peat (88 % of the 1880 wet land) was already wet land in 1878–80, 1880 Moor is today's raised bog, and part of the former wet meadows has lost its peat since.

Step 09 – Atlas (PyQGIS)

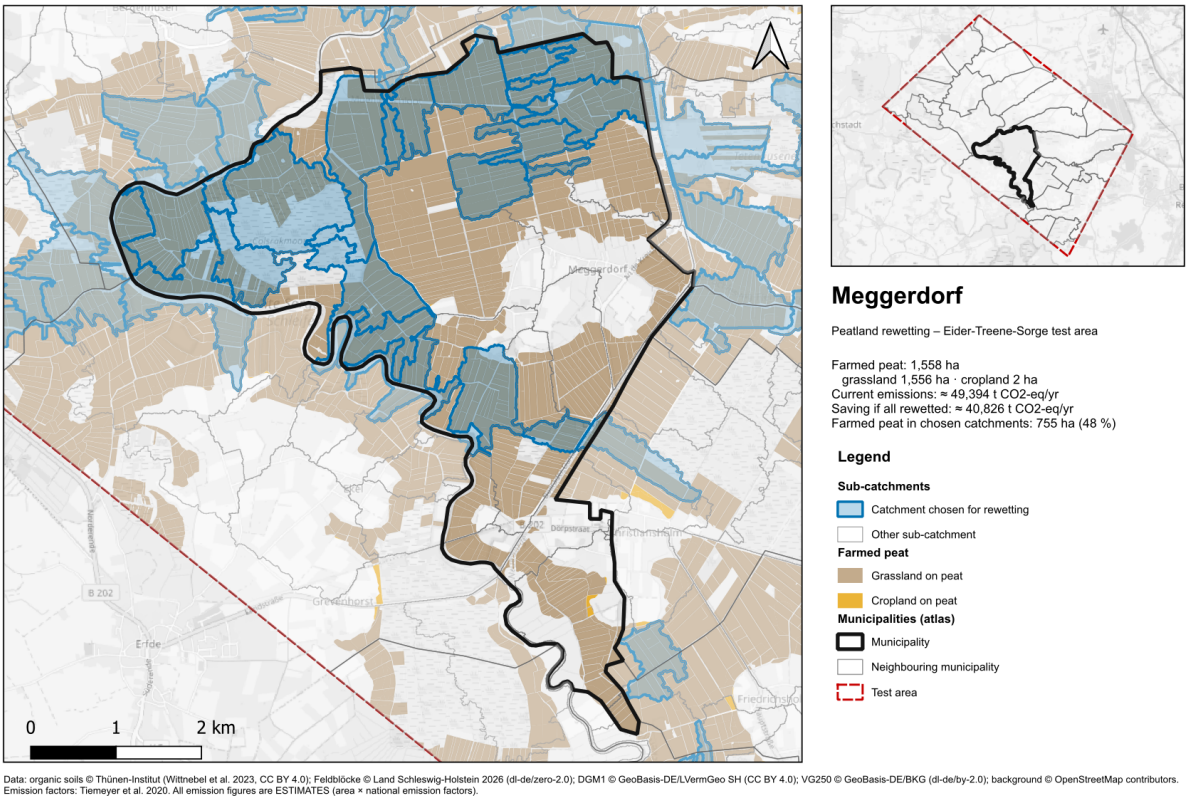
Why

Results have to be communicated per municipality – the level at which planning and talks with landowners happen.

What the script does

- builds a coverage layer: the 24 municipalities with at least 50 ha farmed peat, with key figures
- loads and styles all map layers in a separate group (other layers in the project stay untouched)
- creates the print layout 'Atlas_Gemeinden' (A4 landscape): main map following each municipality, overview map, title, key figures, legend, scale bar, north arrow, sources
- exports one PDF with all pages plus one PNG per municipality, biggest emitters first

Results (test area)



Atlas page for Meggerdorf. Blue: sub-catchments chosen by the catchment optimum; brown: grassland on peat; black: the municipality.

Decisions and why

- Everything scripted, so the maps can be re-made after any analysis change in one run.
- Legend fonts set explicitly (QGIS defaults are far too large for an 84 mm column); English number format; sources in a strip along the page bottom.
- zoomToExtent instead of setExtent – the latter silently resizes the map frame.

Limitations

- The PDF is still ≈ 77 MB (many detailed polygons on 24 pages); for GitHub or e-mail, a reduced version is needed.
- Re-running the script overwrites manual changes in the layout.

5 Main findings

- **Most of the peat is farmed, almost all as grassland.** 70 % of the organic soils in the test area are in Feldblöcke, 97 % of that permanent grassland. The cropland/grassland split therefore explains little; what differs between fields is mainly their water level.
- **The order of magnitude is large.** ≈ 354,000 t CO2-eq/yr from ≈ 11,100 ha – about 32 t per hectare and year, consistent with a grassland-dominated area under the national method.
- **The uncertainty is a data gap, not a statistical problem.** The estimate is stable (±1 %) if fields deviate independently, but could be −60 % / +35 % if the whole region is wetter or drier than the German average. Regional water-level data are the key to a better estimate.
- **Without hydrology, optimisation adds nothing.** With uniform factors, ranking and optimisation choose the same area; any apparent 'priority list' of fields is false precision.
- **With hydrology, the picture changes completely.** Planning by sub-catchment reaches the same target while wetting ≈ 80 times less farmland on mineral soil (52 ha vs 4,296 ha) – concentrated in 82 units that are almost pure peat.
- **The peat has been there for at least 150 years – but it is thinning.** 92 % of the land shown as wet on the 1878–80 map is still organic soil; 1880 'Moor' corresponds to today's raised bog. About a third of the 1880 wet meadows are now Moorfolgeboden, and 721 ha of today's fen were already drained pasture or dry meadow in 1880.
- **The historical map can be read automatically.** A small YOLO model plus rule-based dash detection reproduces my hand digitising on 90 % of the area it was not tuned on ($\kappa = 0.86$) and reaches 90 % ± 6 % accuracy in a blind check across the whole test area. Across 304 km², 88 % of the 1880 wet land is still organic soil today.
- **Data quality matters in flat terrain.** Terrain-model resampling (minimum vs mean) and survey dates 20 years apart visibly affect streams and catchments.

6 Limitations of the whole approach

Limitation	Effect	How it could be addressed
National average emission factors	local water level, peat thickness, management ignored	water-level map (Thünen), field data, response functions
DEM-based catchments in pumped polders	pumps, culverts, pipe drains invisible	pumping-station catchments of the water boards; stream burning
Hand-drawn test area	80 sub-catchments cut at the edge	use a real catchment as AOI
Mixed DGM1 survey years	possible steps at tile edges	map and discuss; check edges
Feldblock ≠ ownership parcel	no statement about owners	ALKIS (not open)
No costs, protected areas, willingness	optimum is purely physical	add constraints / costs
Monte Carlo covers factors only	mapping errors not included	add area uncertainty
Historical comparison: map signatures vs. soil map	'only today' is not a peat gain	keep the categories separate; field data on peat depth
Hand digitising of one 36 km² block	position error, subjectivity	edge tolerance reported; YOLO test for reproducibility
YOLO trained in one block	nasse Wiese over-mapped outside it (user's acc. 0.72)	more labelled tiles from other areas; separate pasture rows from wet-meadow dashes

7 Problems encountered and how they were solved

Problem	Cause	Solution
Inputs reported as MISSING	data sat one or two folders deeper after unzipping	paths adjusted in config.py
ValueError when counting features	empty column list not allowed by pyogrio	count from file header (read_info)
'Error adding field area_ha'	GeoPackage field names are case-insensitive (AREA_HA exists)	drop old field before recomputing
Miniforge installer failed	Smart App Control (Device Guard) blocks unsigned programs	use QGIS's bundled Python
PuLP TypeError (cat=…)	PuLP 4.0 changed its API; CBC solver would be blocked anyway	SciPy milp with HiGHS
Uncertainty looked 'massive'	per-hectare ranges applied as area uncertainty	two-piece normal + two correlation scenarios
Impossible negative emissions	unbounded lower tail of the distribution	floor at CO2-Cmin = −0.93 t C/ha/yr
Only one DGM1 tile per order	deliberate limit of the download client	official bulk-download GeoJSON index
CERTIFICATE_VERIFY_FAILED	QGIS's Python doesn't use the Windows certificate store	truststore package (checking stays on)
r.watershed 'not found'	qgis_process loads plugins separately; LTR command name	qgis_process-qgis-ltr plugins enable grassprovider
Streams across fields	mean resampling erased 1–3 m ditches	minimum resampling
Parcels counted in two catchments	touching boundaries counted as intersecting	assign by interior point
Atlas inset too tall, legend overlapping	setExtent resizes the frame; default legend fonts	zoomToExtent; compact legend
Map symbols misread (heath vs. moor, one meadow class)	1933 legend; 1882 key not accessible	official 1898 key of the same survey series
'Field evidence not found' in QGIS	field had been created as 'evidence ' (trailing space)	script strips field names
New polygons cut to slivers	invisible polygons without class + 'avoid overlap'	catch-all symbol category; reassign or delete
Road strips between 1880 polygons	roads and ditches left out while digitising	gap closing (30 m), sensitivity tested
Tufts too small for YOLO	12 px at 1 m; finest YOLO grid step is 8 px	320 m tiles enlarged to 640 px; less scale augmentation
Recall looked low (mAP50 0.52)	IoU criterion too strict for 16 m boxes; low confidences	point-based check; threshold 0.20; duplicates merged
Tile flags not saved	Field Calculator edit left in an open edit session	save layer edits; script reads the flag from disk
World files not found	GDAL writes .wld for PNG, not .pgw	script accepts both
Drained bog read as heath	few dashes on bog heath	rule 'heath in a moor' from the digitising rules
Dry heath read as Moor (whole area)	tuft base lines counted as dashes; heath rule spread	ignore base lines near tufts; heath only joins a larger moor
Spot-check folder empty	tile file size used to skip empty tiles; PNGs compress too well	select by ink share instead
RasterizeLayer TypeError	data source closed before its layer was used	keep the data source in a variable

8 Next steps

- **Second training round:** label 6–8 tiles outside the block with diagonal peat cuttings and the larger tuft style, retrain with rotation augmentation, and check with a new set of points.
- **Separate wet meadow from pasture rows better** – the main error source outside the block.
- Use the corrected 1880 areas to refine the rewetting priorities (e.g. catchments with former wet meadows that are now Moorfolgeboden).
- Optional: stream burning; pumping-station catchments; protected areas as a constraint; a Monte Carlo of the catchment result; extend the historical block.

9 Glossary

Term	Meaning
Organic soil / peat	Soil with a high share of organic matter, formed under water-logged conditions. Niedermoor = fen (fed by groundwater), Hochmoor = bog (fed by rain), Anmoor = transitional soil with less organic matter.
Feldblock (FLIK)	Agricultural reference parcel used for EU subsidies; a block of farmland bounded by permanent features. Not an ownership parcel.
CO2-equivalent, GWP	Other gases converted into the CO2 amount with the same warming effect over 100 years. IPCC AR4 values used here: CH4 = 25, N2O = 298.
Implied emission factor (IEF)	Average emission per hectare for a land-use category including all gases and its undrained share, as used in the national inventory.
DGM1 / DEM	Digital terrain model with 1 m cells: ground height without buildings and vegetation, from airborne laser scanning.
Resampling (min / mean)	Combining fine cells into coarser ones. Mean averages them; minimum keeps the lowest value, preserving narrow channels.
D8 flow direction	Each cell drains to the steepest of its 8 neighbours.
Flow accumulation	Number of upstream cells draining through each cell; high values mark streams.
Sub-catchment	Area draining to one stream segment; here at least 50 ha.
Polder, pumping station	Low-lying land protected by dykes, drained by ditches; pumps lift the water into the embanked river.
Integer programme (MILP)	Optimisation where decisions are whole numbers – here yes/no (0/1) per parcel or catchment.
HiGHS	Open-source optimisation solver built into SciPy.
Greedy ranking	Take the best-scoring units one by one until the target is reached; not guaranteed optimal.
Monte Carlo	Repeat a calculation many times with randomly drawn inputs to see the spread of results.
Two-piece normal	A distribution with different spreads below and above its peak – fits lopsided ranges.
Atlas (QGIS)	A print layout that repeats for every feature of a coverage layer – here one page per municipality.
qgis_process	Command-line tool that runs QGIS Processing algorithms (incl. GRASS) from scripts.
Preußische Landesaufnahme	Topographic survey of Prussia at 1:25,000 (Messtischblätter); in Schleswig-Holstein surveyed 1877–80.
Bruch, Torfstich	Bruch: old word for swamp/fen, on the map tufts on dashes. Torfstich: peat cutting, drawn as chains of small rectangles.
Hutung, nasse / trockene Wiese	Rough pasture; wet and dry meadow – separate symbols in the 1898 key.
Moorfolgeboden (MF)	Former peat soil: drained and decomposed until only a thin organic layer is left.
YOLO	'You Only Look Once' – an object-detection network that finds and boxes all objects in an image in one pass. Here: Ultralytics YOLO26n, the smallest version.
Precision / recall	Recall: share of the real symbols that were found. Precision: share of the detections that are correct.
mAP50	Mean average precision; a detection counts only if its box overlaps the labelled box by ≥ 50 %.
Cohen's κ (kappa)	Agreement between two maps beyond what chance would give; 0 = chance, 1 = perfect, > 0.8 'almost perfect'.
IoU	Intersection over union: overlap of two areas divided by their combined area.
User's / producer's accuracy	User's: of the area mapped as a class, the share that really is that class. Producer's: of the real area of a class, the share the map finds.

Term	Meaning
Stratified random sample	Random points drawn separately within each mapped class, then weighted by the class area.
EPSG:25832 / 4647	ETRS89 / UTM 32N; 4647 is the same projection with the zone number '32' prefixed to the easting.

10 References

- Tiemeyer B., Freibauer A., Borraz E. A. et al. (2020): A new methodology for organic soils in national greenhouse gas inventories: Data synthesis, derivation and application. Ecological Indicators 109: 105838. https://doi.org/10.1016/j.ecolind.2019.105838
- Wittnebel M., Frank S., Tiemeyer B. (2023): Aktualisierte Kulisse organischer Böden in Deutschland. Thünen Working Paper 212. Data: https://doi.org/10.3220/DATA20230510130443-0
- Landesamt für Vermessung und Geoinformation Schleswig-Holstein: Offene Geobasisdaten (OpenGBD), DGM1; download portal https://geodaten.schleswig-holstein.de/gaialight-sh/_apps/dladownload/
- Land Schleswig-Holstein: Feldblöcke, reference date 1.1.2026; metadata https://gdk.gdi-de.org/geonetwork/srv/api/records/21f67269-780f-4f3c-8f66-03dde27acfe7
- Bundesamt für Kartographie und Geodäsie: Verwaltungsgebiete 1:250 000 (VG250), https://gdz.bkg.bund.de
- Land Schleswig-Holstein (2024): Moor-Klimaschutz (January 2024).
- GRASS GIS r.watershed manual; QGIS documentation (Processing, qgis_process, print layouts / atlas).
- SciPy: scipy.optimize.milp (HiGHS solver).
- Landesamt für Vermessung und Geoinformation SH: Preußische Landesaufnahme (Chronologen), WMS https://dienste.gdi-sh.de/WMS_SH_FD_Chronologen, layer Chronologen_1_1878-1880, CC BY 4.0.
- Ultralytics YOLO (YOLO26), https://docs.ultralytics.com – AGPL-3.0.
- Olofsson P., Foody G. M., Herold M., Stehman S. V., Woodcock C. E., Wulder M. A. (2014): Good practices for estimating area and assessing accuracy of land change. Remote Sensing of Environment 148: 42–57.
- Königlich Preußische Landesaufnahme (1898): Zeichen-Erklärung für die Messtischblätter 1:25 000. Reprint 1988, Landesvermessungsamt NRW; PDF from Geobasis NRW (Bezirksregierung Köln).